Software Requirements Specification

for

University Department Information Management System (UDIIMS)

Version: 1.0

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Table of Contents

1. **Introduction**
    - 1.1 Purpose
    - 1.2 Document Conventions
    - 1.3 Project Scope
2. **Overall Description**
    - 2.1 Product Perspective
    - 2.2 Product Functions
    - 2.3 User Classes and Characteristics
    - 2.4 Operating Environment
    - 2.5 Design and Implementation Constraints
    - 2.6 User Documentation
    - 2.7 Assumptions and Dependencies
3. **System Features**
    - 3.1 **Module 1: Academic Information System for Students**
        - 3.1.1 Student Information Management
        - 3.1.2 Enrolment and Registration
        - 3.1.3 Academic Performance (GPA/Grades)
        - 3.1.4 Program and Course Management
        - 3.1.5 Fee Status View
    - 3.2 **Module 2: Administrative Information System for Departments**
        - 3.2.1 Department Selection & Navigation
        - 3.2.2 Faculty & Professor Management
        - 3.2.3 Departmental Projects & Research
        - 3.2.4 Inventory Management
        - 3.2.5 Departmental Accounts
    - 3.3 **Module 3: Finance Section**
        - 3.3.1 Grant Management
        - 3.3.2 Fund Management (Consultancy)

        - 3.3.3 Fee Collection & Modification

4. **External Interface Requirements**
    - 4.1 User Interfaces
    - 4.2 Hardware Interfaces
    - 4.3 Software Interfaces
    - 4.4 Communications Interfaces
5. **Non-Functional Requirements**
    - 5.1 Performance Requirements
    - 5.2 Safety Requirements
    - 5.3 Security Requirements
    - 5.4 Software Quality Attributes

1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the University Department Information Management System (UDIIMS). It aims to explain the purpose and features of the system, the interfaces of the system, what the system will do, and the constraints under which it must operate.

1.2 Document Conventions

- **Bold** text is used to identify specific UI buttons, database fields or critical emphasis.
- **SRS:** Software Requirements Specification.
- **UDIIMS:** University Department Information Management System.
- **GUI:** Graphical User Interface

1.3 Project Scope

The UDIIMS is a desktop-based software solution designed to automate the manual bookkeeping of university departments. Currently, student records, inventory logs, and financial records are maintained in separate files. This system integrates:

1. **Student Information:** Academic history, personal details, and course registration.
2. **Department Administration:** Managing faculty, specific inventory (Labs vs. Classrooms), and research output.
3. **Finance:** Centralized tracking of grants, consultancy funds, and student fee collection.

The system will be built using **Java** for the application logic and **Supabase** for data persistence.

2. Overall Description

2.1 Product Perspective

The University Department Information Management System (UDIIMS) is designed to automate and integrate the academic and administrative operations of university department offices that are traditionally managed through manual bookkeeping. The system provides centralized management of student records, including admissions, semester-wise course registrations, backlog tracking, grading, and automatic GPA/CGPA calculation, along with generation of grade sheets. Additionally, the system enables efficient data entry by department staff and provides query facilities for retrieving up-to-date student and financial information, thereby improving accuracy, efficiency, and accessibility of departmental records.

2.2 Product Functions

The major functions of the system are:

- **Student Data Management:** Editing personal info, calculating SGPA/CGPA, generating grade cards.
- **Departmental Resource Management:** Tracking professors, publications, and physical assets (inventory).
- **Financial Tracking:** Managing incoming grants, outgoing expenses, and verifying student fee payments.

2.3 User Classes and Characteristics

- **Department Secretary (Administrator):**
  - Access Level: High (Department Specific).
  - Capabilities: Can add/remove courses, enter inventory data, update faculty details.
- **Finance Officer (Administrator):**
  - Access Level: High (Finance Specific).
  - Capabilities: Can modify fee status (Paid/Due), enter grant details, and view financial reports.

2.4 Operating Environment

OE-1: The UDIIMS shall operate as a web-based application accessible through modern web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

OE-2: The UDIIMS backend shall be developed using Java and shall run on a standard application server.

OE-3: The UDIIMS shall use Supabase (PostgreSQL-based backend service) for data storage, authentication, and database management.

2.5 Design and Implementation Constraints

- **Language:** The backend system is developed strictly in Java.
- **GUI:** The user interface shall be web-based and accessible through standard web browsers.
- **Connectivity:** The backend system shall interact with the Supabase database using RESTful APIs provided by Supabase.
- **Downtime:** The database must be available 24/7 for student queries.

2.6 Assumptions and Dependencies

- It is assumed that the database instance is always online.
- It is assumed that "Lab Inventory" is only applicable to science/technical departments (e.g., CSE, Physics) and not humanities (e.g., English, History).

3. System Features

This section outlines the specific behaviours of the system, organized by the three main portals.

3.1 Module 1: Academic Information System

3.1.1 Student Information Management

- **Description:** The entry point for the student module.
- **Input:** Student Roll Number.
- **Processing:** The system queries the Student table. If the Roll Number exists, the dashboard loads. If not, an error "Invalid Roll Number" is displayed.
- **Output:** Student Dashboard with access to Profile, Academic Performance, Courses, and Fee Status.

3.1.2 Enrolment and Registration

- **Description:** Handles semester-wise course registration.
- **Features:**
    - Display of currently enrolled courses.
    - Add/Drop functionality for electives within the registration deadline.
- **Processing:** Updates the Course Registration table and maintains backlog status for failed courses.

3.1.3 Academic Performance (GPA / Grades)

- **Description:** Provides a comprehensive view of the student's academic progress.
- **Data Displayed:**
    - Semester-wise SGPA table.
    - Cumulative CGPA.
    - Option to download grade card.

- **Processing:**
    - SGPA = Σ(Grade Points × Credits) / Σ(Credits)
    - CGPA calculated as cumulative weighted average.
- **Output:** Downloadable grade card in PDF or text format.

3.1.4 Program and Course Management

- **Description:** Tracks academic structure and student progress.
- **Features:**
    - View completed courses.
    - View pending backlogs (courses with grade ‘F’).
    - Summary of total credits earned vs required.

3.1.5 Fee Status View

- **Description:** Displays student fee payment details.
- **Display:** Semester-wise fee table with amount, payment status, and date.
- **Constraint:** Read-only access for department admin; modifications allowed only by Finance staff.

3.2 Module 2: University Department Portal

3.2.1 Department Selection & Navigation

- **Description:** Allows selection of a department context. (e.g., "Department of Computer Science" or "Department of English").
- **Impact:** All administrative data (faculty, inventory, accounts) is filtered based on the selected department. (e.g., WHERE dept_id = 'CSE').

3.2.2 Faculty & Professor Management

- **Description:** Manages faculty records within the department.
- **Data Displayed:**
    - Personal details and designation.
    - Subjects taught.
    - Research interests and publications.
- **Admin Operations:** Add, update, or remove faculty records.

3.2.3 Departmental Projects & Research

- **Description:** Maintains records of departmental research activities.
- **Data Elements:** Project title, supervising faculty, student members, abstract, and publication links.
- **Functional Requirement:** Clicking a publication link opens it in the system's default browser.

3.2.4 Inventory Management

- **Description:** Tracks departmental assets.
- **Features:**
    - Common inventory items (furniture, computers, rooms).
    - Lab-specific inventory for lab-based departments.
- **Operations:** Add, update, and remove inventory items.

3.2.5 Departmental Accounts

- **Description:** Manages internal financial records of the department.
- **Features:**
    - View allocated budget and grants.
    - Track expenses and remaining balance.

3.3 Module 3: Finance Section

3.3.1 Grant Management

- **Description:** Manages grants received by departments.
- **Functions:**
    - View grant history.
    - Record new grants.
    - Track grant utilization.

3.3.2 Fund Management (Consultancy)

- **Description:** Tracks revenue generated through consultancy services.
- **Functions:**
    - Department-wise revenue listing.
    - Entry of new consultancy income records.

3.3.3 Fee Collection & Modification

- **Description:** Administrative control over student fee records.
- **Input:** Student Roll Number.
- **Actions:**
  - View fee status.
  - Mark fees as Paid or Due.
- **Processing:** Updates are immediately reflected in the database and student view.

4. External Interface Requirements

4.1 User Interfaces

- **Style Guide:** The application shall use a tab-based and component-driven web interface implemented using React, allowing users to switch between major views without reloading the page.
- **Forms:** Data entry forms shall implement client-side validation using React (e.g., restricting non-numeric input in phone number fields) and server-side validation in the Java backend.
- **Reports:** Data tables shall support sorting and filtering by columns (e.g., sorting students by GPA).

4.2 Hardware Interfaces

- **Screen Resolution:** Minimum 1024x768 (Standard Desktop).
- **Printer:** The system shall support sending the "Grade Card" and "Fee Receipt" directly to the default OS printer.

4.3 Software Interfaces

- **Database:** The system shall use Supabase (PostgreSQL) for persistent data storage, accessed through Supabase APIs from the Java backend.

5. Non-Functional Requirements

5.1 Performance Requirements

- **Response Time:** Database queries (e.g., searching for a student) must return results within **2 seconds**.
- **Concurrency:** The system must support multiple Department Secretaries entering data simultaneously without data corruption (Database Transaction Locking).

5.2 Security Requirements

- **Access Control:**
  - Finance Section can only be accessed by finance officers.
  - Finance Officers cannot modify Student Grades.
- **Data Privacy:** Data is visible only to individuals of specific user class .

5.3 Reliability

- **Data Integrity:** The system must ensure that the "Funds Remaining" is always calculated as (Opening + Grants + Income) - Expenses. It should never be negative without a warning.